UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS

GCE Ordinary Level

MARK SCHEME for the October/November 2009 question paper for the guidance of teachers

5054 PHYSICS

5054/04 Paper 4 (Alternative to Practical), maximum raw mark 30

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1 **(a)** mass / weight difference between two halves / rule not uniform / C of G or C of M not in centre of rule / not at 50 cm mark / one end (not A) worn away / damaged / holes drilled in one end
(ignore comments on forces / moments / fulcrum / prism) B1 [1]

(b) **(i)** 88.4 ± 0.2 (cm) unit not required B1

(ii) 38.4 ± 0.2 cm OR **(b) (i)** – 50 correct unit required B1 [2]

(c) axes: correct way round, labelled quantity and unit
minimum two values (excluding 0,0) on each axis B1

scales: more than ½ page, sensible, linear
x-axis: 2 cm = 100g or 80g y-axis: 2 cm = 10 cm or 8 cm B1

5 given points plotted accurately, ± ½ smallest square
(for sensible scales only) / ignore point at 50g B1

reasonable attempt at best fit curve neatly drawn
from 75g to 400g, not going below 4cm / ignore line outside points B1 [4]

(d) **(i)** points on graph should be evenly spaced/uniform / large gap (between 200g and 400g) explained somehow
allow more accurate with reference to graph / curve / results / experiment / conclusion / allow clearer graph / fewer gaps in graph B1 [1]

(ii) 6.5 $\pm$ 0.5 cm or answer correct from their graph correct unit required
allow 6-7 cm / allow even if scale not sensible B1

marking clearly shown on graph at their value
minimum requirement: clear mark on curve at 300g B1 [2]

(e) d > 50 cm / metre rule not long enough
allow distance too large/ too close to end of ruler / value of d too high
ignore distance large / cannot balance ruler / comments relating to graph B1 [1]

[Total: 11]

2 **(a)** trundle wheel / measuring tape / more than one metre rule / metre tape
ignore 10 m ruler B1 [1]

(b) know when to start stopwatch / when sound is made / when pistol fired B1 [1]

(c) avoid echoes / reflections / sound bouncing off buildings
ignore noise from surroundings / sound hits buildings B1 [1]

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(d) mark 1 and 2 together any TWO sensible suggestions, e.g.

reactions too slow

ignore just human error

stopwatch faulty / failed to start / stop / not at zero

time interval too short / speed too fast / distance too short

poor hearing / could not hear the bang

poor eyesight / could not see the smoke

different distances (from teacher)

lack of concentration / not paying attention B2 [2]
ignore faulty pistol

NOT parallax error

(e) faster / larger / speed quicker B1 [1]
ignore takes less time / hears sound quicker

[Total: 6]

3 **(a)** uses light from a distant object B1
allow parallel light mentioned / at least 2 parallel rays drawn (not including axis unless arrow shown)

moves lens/focuses light onto a screen B1
allow f/F/15 correctly marked on ray diagram

measures from lens to screen / image / repeats for all 3 lenses B1 [3]
max 2 if concave lens drawn

(b) N increases, f decreases / inverse relationship C1

inversely proportional / N x f = constant A1 [2]
allow max 1 mark if focal point used for focal length

allow one mark if inversely proportional stated then incorrect description

[Total: 5]

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4 **(a)** circuit containing power supply, wire, resistor (allow variable R), (switch) **and**

ammeter drawn in series with wire B1
voltmeter drawn in parallel with wire B1 [2]

allow any representation clearly ammeter and voltmeter / meters drawn on Fig 4.1
ignore gaps in circuit
max 1 if resistor drawn as fuse

(b) makes current small / stops wire heating up / overheating / melting B1 [1]
allow controls / limits current / protects ammeter
ignore resists the current / varies the current / no damage to wire / comments on safety

(c) **(i)** 1.4 V cao unit required B1

(ii) 78 mA / .078 A cao unit required B1 [2]

(d) 0.33 (0.326) (mm) ecf **(c) (i)** and **(ii)** B1

Their answer correctly rounded to 2 s.f. B1 [2]

ecf 78 A in **(c) (ii)** gives 10(.3) mm so 10 scores 2 marks

(e) micrometer / screw gauge B1 [1]
allow caliper ecf (d) ≥ 9 mm

[Total: 8]

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